A Short Survey and Brief Tutorial on the State of the Art in Internet Routing Security

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***Abstract*—This paper examines recent advancements in securing Internet routing, focusing on Resource Public Key Infrastructure/Route Origin Validation (RPKI/ROV) and revisiting established mechanisms like Internet Routing Registries (IRRs). We begin by briefly outlining Internet routing, with a particular emphasis on the Border Gateway Protocol (BGP). Following this introduction, we categorize common routing incidents, including leaks and hijacks. Finally, we delve into the techniques and protocols designed to mitigate these security threats, detailing the scenarios for which they are most suitable.**

I. Introduction

The increasing frequency of routing security incidents has driven significant interest in securing the Internet's routing infrastructure. This article reviews several of these incidents and their impact on the stability and integrity of Internet services. These issues are gaining prominence within the Internet Corporation, spurring improvements over traditional technologies. This document details advancements in routing security, encompassing Resource Public Key Infrastructure/Route Origin Validation (RPKI/ROV), Border Gateway Protocol Security (BGPSec), and Autonomous System Provider Authorization (ASPA). Concurrently, it reexamines enduring mechanisms such as Internet Routing Registries (IRRs), noting their recent resurgence in interest. Finally, the recommendation for the optimal application of each technique is presented. This document is organized as follows: Section II presents fundamental concepts about the organization and operation of the Internet, particularly about its dependence on the BPG protocol; Section III examines the practical consequences of certain BGP protocol flaws on applications integrity and users security; Section IV reviews most typical types of BGP security incidents; Section V outlines enhancements to existing technologies aimed at preventing or mitigating the impact of such incidents. Finally, Section VII provides a summary of the concepts explored in this article and highlights current initiatives and emerging trends within the Internet community's ongoing efforts.

II. About Internet Routing

This section summarizes the fundamental concepts underlying this work, from technological to organizationals.

A. Forwarding and Routing

Packet networks rely upon two distinct but related functions, namely *packet forwarding* and *routing*. When a network device receives an IP (Internet Protocol) packet, it must determine the appropriate outgoing port or interface to forward the packet towards its destination. This *forwarding decision* is made by consulting a pre-populated forwarding table, which maps destination IP addresses to outgoing interfaces.

The *routing function* is responsible for populating this table, using various routing protocols. These protocols can be broadly categorized as *internal* or *external*. Internal routing protocols or Interior Gateway Protocol (IGP) operate within networks under a single administrative control, discovering and maintaining detailed knowledge of the network's topology. External routing protocols are designed for inter-domain routing, connecting large networks to the internet. The Border Gateway Protocol (BGP) is widely recognized as the de-facto standard for external routing protocols on the Internet. While BGP lacks detailed topological information and fast convergence, its superior scalability and ability to clearly mark administrative boundaries make it ideal for this crucial for this role, surpassing the capabilities of internal routing protocols to disseminate global Internet addresses.

B. Addressing, Labeling and Other Protocol Parameters

The Internet is divided into disjoint administrative zones or Autonomous Systems (ASes). ASes are identified by 32-bit numbers usually denoted AS-IDs. Routing protocols populate forwarding tables to map from IP addresses to outgoing interfaces, based on the assumption that each address corresponds to a specific destination (i.e., an AS). During inter-domain routing, the BGP protocol advertises blocks of contiguous addresses associated with the originating AS-ID. These AS identifiers must also be unique, globally assigned numbers.

The Internet Engineering Task Force (IETF) hosts the process through which new Internet protocols are discussed, created, maintained and also sometimes deprecated. This process produces documents known as Requests For Comments (RFCs). Some protocol parameters that need to be *globally unique* for the protocols to operate correctly. The *Internet Assigned Numbers Authority* is responsible for coordinating

several *protocol parameter* databases [1], such as IP addresses, AS numbers, the root DNS (Domain Name System) zone, well-known TCP port numbers, SNMP OIDs, are examples of these registries.

Since 1998, the *Internet Corporation for Assigned Names and Numbers* hosts the IANA functions, which until 2016 ICANN performed under contract with the United States government. From 2016 onwards IANA became a non-profit subsidiary of ICANN performing the mentioned registry functions on behalf of the Internet community.

Further down from IANA, the allocation and management are geographically delegated to Regional Internet Registries (RIRs), which include: the African Network Information Centre (AFRINIC), the Asia Pacific Network Information Centre (APNIC), the American Registry for Internet Numbers (ARIN), the Latin America and Caribbean Network Information Centre (LACNIC), and the Réseaux IP Européens Network Coordination Centre (RIPE NCC). IANA partitions the IPv4 (or IPv6) address space in large blocks, which are *assigned* to the RIRs. The RIRs themselves proceed in the same manner, dividing the blocks received from IANA and assigning them to their own *members*. All five RIRs are legally constituted as non-profit, membership-based organizations.

BGP works by establishing TCP connections between neighbors known as *BGP peers*, each running an instance of the so called *BGP process*, which maintains its own *Routing Information Base* (RIB). BGP peers are also called *adjacent routers* in the BGP overlay. This RIB serves as the foundation for the router to create and maintain its final routing and forwarding tables, utilizing a specific set of rules for the routing: the *BGP decision-process*, or *BGP path-selection*.

BGP peers exchange routing information via BGP messages. The UPDATE message serves to *announce* or *withdraw* routes, carrying not only IP prefixes but also a rich set of BGP *attributes*. These attributes allow network administrators to implement sophisticated *routing policies*. A comprehensive description of all BGP attributes is beyond the scope of this paper. Sections 5 and 9 of [1] detail the most important path attributes and their influence on BGP's route selection process. To get to a more accessible introduction to BGP features, see Cloudflare's website [2].

This article highlights two key BGP attributes: *Next-Hop*, the IP address of the next router; and *AS-Path*: the sequence of Autonomous Systems an update traverses. When BGP updates arrive at routers (e.g., `Router-A` and `Router-B` in `AS-X`, Fig. 1) from different neighboring ASes (`AS-Y` and `AS-Z`) but sharing the same origin (`AS-M`), they are treated as external BGP (eBGP) updates. These updates are forwarded only if selected as the best route. Internally, routers like *Router-C* receive these updates via internal BGP (iBGP). Due to the *split-horizon rule*, iBGP updates aren't forwarded to other internal routers unless route reflection is used [3], preventing routing loops. The AS-Path helps detect and prevent eBGP routing loops: *updates containing the local AS are discarded*. It also aids in route optimization: *shorter AS-Paths are preferred when other attributes are equal*. Finally, Next-Hop determines how routers select routes. The chosen route must be reachable, which is ensured by the Interior Gateway Protocol (IGP). Sub-optimal route selection choices can arise in BGP due to sometimes difficult implementation of coordination between IGPs and also due to the use of route reflectors in large scale networks [4].

[1]These databases are known as *registries* in IANA parlance

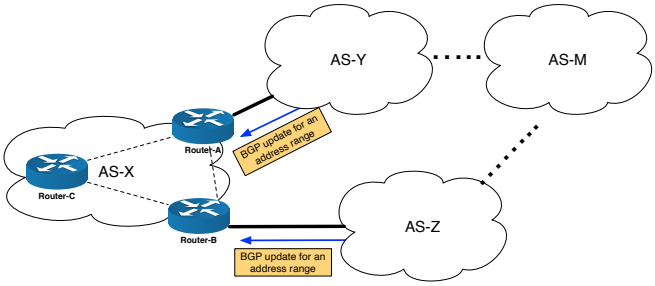


Fig. 1. Schematic example for an incoming eBGP update

It is important to note that, regardless of the process, once entries are installed in the forwarding table, they consist of `<address range, next-hop>` pairs, which may overlap. When a destination IP matches multiple address ranges, the entry with the smallest range (i.e., the most specific) is the one to be selected.

C. Assigning names to IP addresses

The *Domain Name System* provides a mechanism that allows referring to an IP address by name. These names are grouped in *domains*. Each domain constitutes what is called a *zone*. A zone in turn is a collection of *records*. Each record represents a mapping between an attribute/type of a DNS name and a value. A specification of the standard –including most common DNS record types– is in [5].

The DNS is structured as an inverted tree, where the top-level root zone provides referrals to the next level called TLDs or *top level domains*. Queries to resolve a domain name are conceived as an iterative process. DNS servers aggressively cache intermediate responses in order to provide the best response times to clients.

The management of the top of the inverted tree, known as the *root zone* is also delegated to IANA.

III. The Importance of Routing Security

The fundamental goal of network routing is to provide routers with the information necessary to build accurate, complete, and, above all, trustworthy forwarding tables which in turn ensures the network correctly delivers IP packetes to their desinations.

BGP routing is built on *mutual trust*: "each speaker assumes received information from peers is valid". This model, suitable for the early, smaller Internet community, struggles under today's vast mesh of Internet autonomous systems run by thousands of ISPs. Similar trust-based vulnerabilities exist in other systems, like the SS7 telephony protocol, which impacts

mobile services [6], [7]. BGP's flexibility enables scale and while powerful, also becomes a weak point and a vulnerability.

The motivations behind such attacks vary. They can result in denial-of-service incidents that block legitimate traffic from reaching its destination.

Furthermore, employing route hijacking for traffic redirection for inspection and subsequent reinjection into the Internet is a distinct possibility. Since operators seldom monitor AS-paths, these type of attacks often goes unnoticed by the communicating endpoints.

What makes the situation even more challenging is that not all BGP issues stem from malicious activities; many routing integrity problems arise due to accidental misconfigurations.

We can classify the motivations as follows:

- Malicious Intent:
  - Hijacking: Redirecting traffic for data theft, eavesdropping, or launching further attacks like phishing.
  - Denial of Service (DoS): Flooding BGP resources to disrupt routing or blackholing traffic.
  - Creating Instability: Disrupting network connectivity and causing widespread outages.
  - Cyber Espionage: Compromising routers to sniff traffic or gain access to sensitive information.
- Accidental Errors:
  - Misconfigurations: Errors in router settings or policies causing route leaks or false advertisements.
  - Hardware/Software Issues: Failures or bugs triggering route flaps or session drops.
- Traffic Engineering: Traffic Engineering in Internet service providers is understood as the practice of optimizing network link usage. Left to its own devices, BGP will choose a single best path which can lead to situations where one link is heavily used while other equivalent links remain underutilized. The typical way of performing TE in ISPs is either via prefix deaggregation and/or applying AS path prepends. While usually this practice goes unnoticed, misconfigurations in traffic engineering can inadvertently cause severe disruptions.

A classic example of a routing security incident is the 2008 `YouTube` traffic redirection by Pakistan Telecom [8]. This incident was not the first but it has been extensively analyzed over the years.

On February 24th, 2008, Pakistan Telecom inadvertently announced `YouTube`'s IPv4 address space via BGP to its peers. Trusting this announcement, these peers propagated the route, causing all `YouTube` traffic from Central Asia to be directed toward Pakistan Telecom's network.

Quickly overwhelmed by the traffic volume, Pakistan Telecom's network suffered a denial-of-service event, simultaneously impacting its own customers and `YouTube` viewers across the region.

Below are some significant applications and incidents that have arisen from the misusage and outright abuse of BGP.

A. BGP and Content Delivery Networks (CDNs)

Content Delivery Networks (CDNs) are distributed caches that speed up web content delivery by serving data from servers nearest to users. They reduce latency, ease the load on origin servers, and boost availability. CDNs help scale services globally, manage traffic spikes, and providing resilience against distributed denial-of-service (DDoS) attacks.

For user redirection, CDNs leverage the flexibility of DNS and BGP protocols. The DNS is a key component in the CDNs architecture.

This mechanism assumes nearby cache servers are widely available. Indeed, most medium to large ISPs operate data-centers hosting cache servers for global platforms. According to reports [9] and [10], when combining traffic from CDNs like `YouTube`, `Facebook`, `Netflix`, and `Akamai`, an ISP can serve over 70% of its customers' traffic from its data-centers, by means of a mix of cache servers of the previous platforms. This not only improves the user experience, but also reduces traffic on long-distance/high-cost ISP network links. To serve CDN content exclusively to users of an ISP, eBGP adjacencies are set-up between the cache's local server farm and the ISP's AS. The CDN backend aligns global DNS responses with local BGP updates, announcing address ranges to match DNS replies.

In Fig. 2 we illustrate the interaction between DNS and BGP in a CDN context. Three edge caches each serving a different region are shown. Users query their local DNS resolver, which returns the IP address of the nearest edge cache.

Consider the content of `YouTube`. The company must advertise its IP addresses globally via BGP to ensure worldwide access to its content. Currently this range includes, among others, the IPv4 range from 216.58.192.0 to 216.58.223.255, namely 216.58.192.0/19.

In addition, local caches announce (via local eBGP peerings) more specific subranges (e.g., 216.58.192.0/24). Later, these more specific addresses are returned by `YouTube`'s authoritative DNS server to the ISP's recursive one.

Since routing favors more specific prefixes, content is to be served locally as long as the cache is active. Even if the cache fails, routing mechanisms automatically redirect traffic to a more distant source with a less specific prefix.

If the ISP relays these BGP updates (locally received) to other ASs, it could make content hosted in its data-center available to users other than its own, potentially causing security incidents, such as the one described below.

Content Delivery Networks (CDNs) help mitigate routing security incidents primarily by localizing traffic and providing a resilient overlay that is less sensitive to erroneous or malicious BGP announcements. By caching content on topologically distributed edge servers, CDNs steer user requests toward nearby nodes, reducing the dependency on long interdomain paths— in effect, minimizing the surface exposed to route leaks or hijacks and limiting the impact of any single BGP misannouncement.

CDNs and cloud providers exchange traffic with thousands of networks and often participate in initiatives such as the

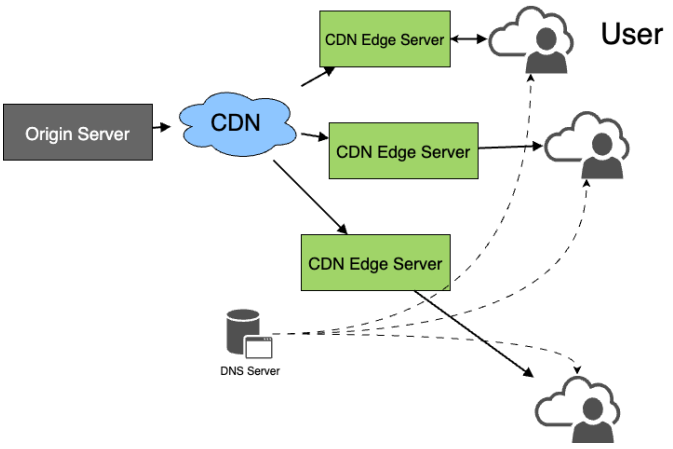


Fig. 2. Schematic example of a CDN.

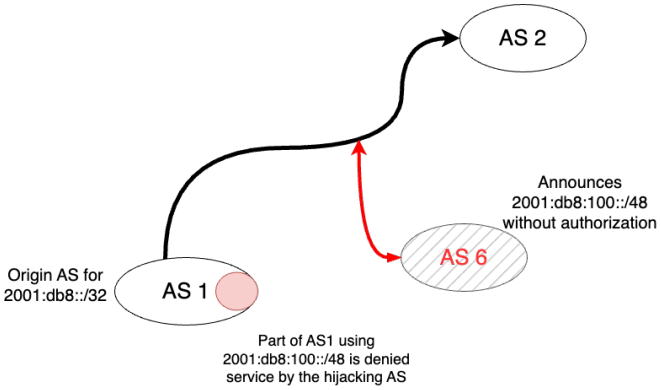


Fig. 3. A route hijack scenario

MANRS CDN and Cloud Provider Program [11], they are encouraged to implement strict filtering, validate peering routes, and rapidly reroute or absorb surges caused by malicious or misconfigured announcements, thereby enhancing overall Internet routing hygiene.

The 2008 Pakistan Telecom / Youtube incident occured in pre-CDN days. Had this way of publishing content been in place, the incident might have been mitigated and would probably would have affected only a small subset of users.

IV. Taxonomy of Routing Security Incidents

Routing security incidents can be broadly classified into *route hijackings* and *route leaks*, which are described in the following paragraphs.

A. Route Hijacks

An Autonomous System (AS) that originates a route acts as the *home* network for its assigned clients, announcing their IP prefixes via BGP. Upstream networks then propagate these announcements, allowing others to reach those clients. A route hijack occurs when an upstream network wrongly announces IP prefixes it doesn't own, conflicting with the legitimate origin AS, and receiving traffic in its behalf. This can result from malicious intent or simple misconfiguration –such as a typo– especially if no safeguards are in place.

The hijacker typically doesn't deliver traffic to the intended destination, dropping it within their network thus cause a service disruption affecting the rightful originator, creating a denial-of-service kind of scenario. The 2008 Pakistan Telecom incident described in [8] is a very well known example.

However, in some malicious cases the hijacker can re-inject this redirected traffic into the Internet in a way that the affected users notice little or no changes in their service unless they actively monitor their traffic's path across the Internet. This allows the attacker for example to inspect traffic looking for sensitive user information without alerting them.

Fig. 3 illustrates a route hijack scenario.

`AS1` has the legitimate right to use the IPv6 prefix `2001:db8::/32`. However, at some point between `AS1` and `AS2`, `AS6` announces `2001:db8:100::/48`, a more specific portion of whole prefix.

Hence, `AS6` "hijacks" (i.e., creates a redirection of) a portion of the traffic from `AS2` destined for `AS1`, disrupting service for some of `AS1`'s users, since this traffic never reaches its intended destination. [2]

B. Route Leaks

Route leaks, while related to hijacks, represent a different type of events. As defined in [12], a route leak occurs when a BGP routing announcement propagates beyond its intended scope.

While hijacks typically involve a small number of prefixes, leaks often involve thousands, leading to significantly broader disruptions. Kentik describes in [13] two such large-scale route leak incidents that caused major Internet traffic disruptions.

Fig. 4 shows a common route leak scenario involving two autonomous systems, `AS1` and `AS2`, which are connected to the Internet and also maintain a private BGP peering for direct traffic between them –not meant for general Internet transit–.

If the smaller `AS2` mistakenly leaks Internet routes to `AS1`, it might look as a better path to the Internet, and some of `AS1`'s routers then route traffic through `AS2`. However, `AS2` often lacks the capacity or policy permissions to handle this unexpected traffic, resulting in service disruptions.

Detecting and preventing route leaks is significantly more challenging than identifying route hijacks. While hijacks are easier to spot due to changes in the origin AS, route leaks retain the correct origin, making them less obvious. Although the AS path changes significantly, it still appears valid from the control plane's perspective.

The authors of [14] propose a novel tool to validate new paths, which consists in assigning *roles* to the different BGP sessions established in a neighbor. For example sessions can be of role "Provider" or role "Peering". Prefixes learnt through

[2]Bear in mind that the users using the 2001:db8::/32 range are in AS1, traffic going from AS1 to AS2 destination addresses in AS2's announced ranges and thus is unaffected by the spurious AS6 announcement. However, return traffic from AS2 to AS1 has destination addresses in the 2001:db8::/32 range and is affected by the hijack.

"Peering" sessions should not include Aces not part of the peering in the AS path.

C. IP and ASN Squatting

IP squatting refers to the unauthorized, opportunistic use of public IP address space –outside RFC 1918 ranges [15] or IPv6 ULA prefixes (fc00::/7)– typically to quickly gain network resources and evade detection. Malicious actors announce IP prefixes via BGP without legitimate rights to these addresses, which often come from legacy allocations[3] or unassigned blocks that Regional Internet Registries (RIRs) no longer actively monitor.

In a similar way AS numbers can also be squatted, resulting in announcements seemingly coming from the correct ASN but in practice are false and could be used to bypass some of the countermeasures described in the rest of this document.

Routers have no inherent way to verify if an IP prefix announced via BGP is legitimately authorized. Transit and peering networks enforce these rights using tools like: route maps, prefix lists, or ACLs (access controls lists). Due to BGP's trust-based nature, attackers can announce any prefix from their network. Without proper validation by upstream providers, these squatted prefixes become routable on the public Internet. See [16] for a detailed discussion on who and why they squat on IPv4 prefixes.

Squatted IP addresses often host malicious services like spam servers, command-and-control nodes, or phishing websites, frequently bypassing filters and ACLs due to their lack of known previous use. Tracing and attribution become challenging as usually the squatted space is not registered.

In [17] a routing incident arising from a massive IP squatting operation is described.

V. Strategies for Improving Routing Security

To combat prevalent routing threats, the Mutually Agreed Norms for Routing Security (MANRS), backed by the Global Cyber Alliance, establishes critical security measures. These measures are implemented through four distinct programs tailored for Network Operators [18] and Internet Exchange Points [19]. It also includes resources for CDN and Cloud Providers, and Equipment Vendors. In the case of Network Operators the key measures are:

- Filtering: Preventing the propagation of incorrect routing information by ensuring the correctness of their own announcements and announcements from their customers.
- Anti-spoofing: Preventing traffic with spoofed source IP addresses by enabling source address validation.
- Coordination: Facilitating global operational communication and coordination among network operators.
- Global Validation (with IRR): Facilitating routing information on a global scale through the Internet Routing Registry (IRR).
- Global Validation (with RPKI): Facilitating routing information on a global scale through the Resource Public Key Infrastructure (RPKI)

We now turn our attention to techniques for enhancing routing security, starting with the most widely deployed and progressing to two newer approaches.

A. Internet Routing Registries

Internet Routing Registries (IRRs) are distributed, global databases where network operators, typically those using BGP, create entries, known as *objects*, that represent their intended routing *policies* [20]. These entries are defined using the Routing Policy Specification Language (RPSL) [21].

Since routers' operating systems usually lack advanced text processing abilities, they cannot directly parse RPSL policies. Instead, the content of the IRR databases is pre-processed using tools such as **bgpq3** [22] or **bgpq4** [23].

These tools generate configuration files that when uploaded to routers enable filtering enforcement based on documented routing policies for each AS. Both support multiple hardware vendors and open source BGP implementations.

IRRs have been part of the Internet landscape since its early days, and enjoyed widespread use for some time. Their use declined due to two main issues: weak authentication and lack of secure transport for RPSL data. Many IRRs still depend on basic password or email-based verification, making them vulnerable to abuse. See [24] for a survey on invalid IRR data.

The absence of a central authority for IRRs has led to many poorly managed registries that publish outdated or inaccurate information.

In recent years, IRR usage has rebounded thanks to operational improvements and efforts to clean up stale or invalid data. Although RIRs have long offered IRR services, only recently have these been properly maintained. The number of IRRs has also consolidated into a smaller, more trusted set. This renewed use has drawn attention from attackers. As noted in [24], adversaries have started registering false IRR records to bypass routing defenses during attacks like BGP hijacks.

[3]Address allocations made before the existence of the RIRs which are not under a contract and thus RIR policies are not enforceable on them.

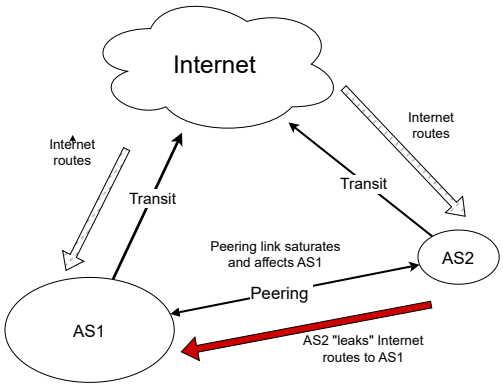


Fig. 4. A route leak scenario

B. Resource Public Key Infrastructure (RPKI) and Route Origin Validation (ROV)

RFC 6480 [25] and RFC 6811 [26] outline a secure Internet routing architecture based on the Resource Public Key Infrastructure (RPKI).

It leverages the hierarchical allocation of IP addresses and AS numbers and defines a distributed repository system for storing and sharing RPKI data and other signed routing security objects.

The key components of this architecture are:

- Resource Public Key Infrastructure: RPKI represents the hierarchical allocation of IP addresses and AS numbers. It uses X.509 certificates to attest to these resource allocations, although it does not include identity verification. RPKI includes Certification Authority (CA) certificates for resource holders and End-Entity (EE) certificates used to validate signed routing objects, such as Route Origination Authorizations (ROAs).
- Route Origin Authorization objects: ROA objects allow IP address holders to authorize specific Autonomous Systems (ASes) to originate routes for their address space. ROAs provide verifiable authorization, which is crucial for the secure construction of BGP route filters.
- Distributed Repository System: It stores ROAs, along with the other components of the Public Key Infrastructure (PKI), such as certificates and Certificate Revocation Lists (CRLs). Its distributed design ensures global availability and integrity of these signed objects.

A Route Origin Authorization (ROA) is a digitally signed statement authorizing an AS to originate specific IP prefixes. The ROA object is part of a PKI structure, enabling full signature validation back to a trusted root (a *trust anchor*) while reducing the number of trust anchors required.

The hierarchical structure of IP address allocation aligns well with the RPKI tree, enabling Regional Internet Registries to provide strong authentication and authorization for ROA management. This cryptographic approach addresses key IRR weaknesses. However, processing ROAs and PKI objects is resource-intensive for routers, so external tools like Routinator [27], FORT [28], and rpki-client [29] are used to fetch and validate data. These tools produce prefix lists that routers use to guide BGP decisions. Routers apply the *Route Origin Validation* algorithm ([26]) and can subsequently drop routes deemed invalid.

RPKI trust is rooted in the RIRs, which already manage IP and AS number allocations. Each RIR maintains an authoritative database, ensuring global uniqueness and linking resources to verified legal entities. To support this, RIRs rigorously authenticate organizations and validate the credentials of personnel authorized to manage ROAs and other RPKI objects.

The RPKI trust model mirrors the X.509-based Web PKI. Just as Certificate Authorities verify identities (e.g., domain validation, organizational checks) before issuing TLS certificates, RIRs validate resource holders before issuing RPKI certificates and signing ROAs. In both systems, a trusted third party enhances security by providing verifiable proof of authority.

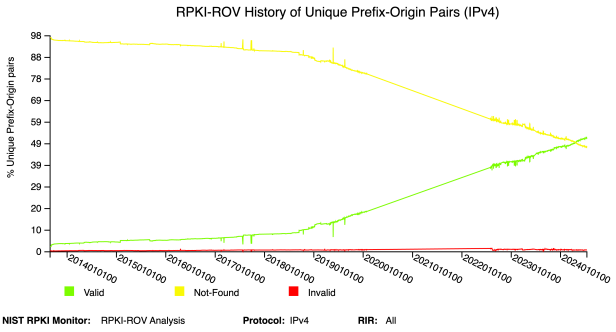


Fig. 5. RPKI Deployment Stats, source [30]

ROAs are less expressive than RPSL objects, as they only authorize origin ASes and cannot validate full BGP paths or upstream/downstream relationships. However, their robust authentication and authorization mechanisms, along with the use of cryptography, have made RPKI the second most widely deployed routing security mechanism after IRRs. ROA coverage now includes about half of all Internet routes, with steady growth over the past five to six years (see Fig. 5).

The NIST SP 800-189r1 report [31] outlines recommended technologies for securing BGP, including RPKI, ROAs, ROA-based origin validation (ROA-ROV), and prefix filtering. To mitigate DDoS attacks, it suggests anti-spoofing methods like source address validation (SAV) using ACLs and unicast Reverse Path Forwarding (uRPF). Additional measures include Remotely Triggered Black Hole (RTBH) filtering, Flowspec, and other techniques also supported by MANRS [18].

C. BGPsec and ASPA

While IRRs and RPKI are currently the most widely deployed techniques for detecting, preventing, and mitigating routing security incidents, they have limitations. Notably, they lack the ability to validate the entire BGP path (beyond the originating AS) within a robust PKI framework. BGPsec [34] extends the BGP protocol, defining a new attribute, BGPsec_PATH, and new capabilities. The BGPsec_PATH attribute, which complements the existing AS_PATH, provides a digitally signed record of the path an announcement has traversed. These signatures are generated by the routers themselves using *router certificates*, which can be validated within the RPKI framework described earlier. While BGPsec was finalized in 2017, its adoption has been limited due to its natural complexity.

A more recent approach, Autonomous System Provider Authorizations (ASPA), is currently under discussion within the IETF [35]. ASPA aims to address the on-router signature verification challenge by introducing an additional object to the RPKI and employing a different validation model. Rather than providing a signed path, ASPA offers a simple valid/invalid status for routes. However, it offers stronger protection against route leaks, an area where BGPsec faces

TABLE I
ROUTING-SECURITY-IMPROVING TECHNIQUES, THEIR PRIMARY USE CASES, IMPACTED PARTIES, DEPLOYMENT STATUS, AND KEY CITATIONS.

Technique	Best Suited for	Roles Benefiting Most	Deployment Status	Relevant References
IRR	Route hijacks and policy enforcement (import/export filtering)	Network operators, ISPs	Widely deployed	[21], [32]
RPKI / ROV	Route hijacks and leaks where origin AS is altered or prefix lengths are misconfigured	Origin AS operators, downstream customers	Significant deployment, growing adoption	[25], [26], [30], [13]
ASPA	Path-based attacks (AS path spoofing) and route leaks	Transit providers, peering partners	Minimal deployment, tooling not yet available for most platforms	[14], [33]
BGPsec	Path-based attacks (AS path spoofing) and route leaks	Transit providers, peering partners	Minimal deployment, tooling not yet available for most platforms	[34]
RFC9234 (Roles in BGP)	Route leak prevention in customer-to-provider sessions	IXPs, large ISPs	Minimal deployment	[14]

challenges. ASPA together with Only to Customer (OTC) RFC9234 [14] provides strong protection against route leaks, and against some forms of AS path manipulation.

ASPA is not yet a standard protocol, and not fully functional, but deployable implementations exist at present. Some research [33] suggests that ASPA can be effective even without full adoption by Tier-1 ASes.

While significant progress has been made, deployment faces substantial real-world obstacles. BGPsec demands router upgrades capable of sustaining cryptographic validation at scale, which is cost-prohibitive for many operators. ASPA is not yet universally implemented across platforms, limiting early adoption. In the case of RPKI, while RPKI requires careful certificate lifecycle management and repository publication, tooling has improved to the point where most of this tasks can be automated.

Moreover, large networks are often risk-averse to altering BGP configurations that already underpin critical services, and the security benefits are difficult to quantify in immediate economic terms. These challenges illustrate that coordination, incentives, and vendor support are as decisive as protocol design in shaping the actual uptake of routing security mechanisms.

VI. FUTURE RESEARCH

Looking forward, several research directions remain open. Achieving scalable path validation with acceptable router overhead is still unresolved, with the trade-offs between BGPsec and ASPA requiring further exploration. Operator usability of cryptographic mechanisms, including automation of certificate management and mitigation of human error as well as algorithm-agility mechanisms, is another critical area. Equally important are empirical studies assessing the effectiveness of initiatives such as MANRS and the broader economic incentives that drive or hinder deployment.

These questions highlight that future work must integrate technical, operational, and economic perspectives if Internet routing security is to advance beyond its current partial adoption.

VII. CONCLUSIONS

Table I summarizes the routing security techniques discussed in this paper, along with the types of incidents they are best suited to address.

An operator must balance security guarantees and deployability. IRRs remain widely used because of their simplicity and expressiveness, yet their weak authentication undermines trust. RPKI with Route Origin Validation offers stronger cryptographic assurances and has reached significant adoption, but its scope is limited to validating the origin AS. BGPsec provides the strongest security model by extending validation to the entire path, but its computational burden has deterred real-world deployment. ASPA represents a pragmatic middle ground, strengthening protection against leaks while retaining operational feasibility, although tooling and vendor support are still emerging. In real-world scenarios operational preference often tilts toward mechanisms that balance ease of deployment with incremental security improvements rather than those offering comprehensive but heavy solutions.

IRRs remain the most widely adopted routing security measure because of their simplicity, lack of cryptographic complexity (e.g., RPKI/ASPA/BGPsec), and the expressiveness of RPSL. This method require no changes to BGP's route selection algorithm and minimal operating system support from routers. Their functionality relies on generating routing configuration statements. However, these very characteristics that make them easy to understand also represent their main weaknesses. The long-standing lack of proper authentication and authorization for IRR updates highlights the need for more secure alternatives. RPKI and ROV introduce proper authentication and authorization tied to the number registries operated by the RIRs. The RIRs verify the legal status of IP address holders, meticulously record transfers, and accurately attribute ownership.

RPKI also adds the comprehensive cryptographic validation operations available in any PKI.

ROV leverages RPKI ROA objects and a small addition to BGP's route selection algorithm to help routers validate the origin AS of each route. However, ROV is less expressive than RPSL objects allowing only for origin validation.

RPKI is also widely deployed; more than 50% of all routes appearing in the Internet's routing table are already covered by ROAs.

BGPsec and ASPA extend RPKI in different ways trying to close the gap left in ROV, particularly concerning the lack of

full path validation. BGPsec aims for a deterministic outcome in full path validation, whereas ASPA proposes an incremental validation approach that considers the diverse nature of BGP interconnects (peering, customer, transit). BGPsec has been fully specified for several years but has not achieved widespread deployment. ASPA can be deployed incrementally, something that could motivate deployment but the standardization process has not yet concluded.

BGP Roles, as defined in RFC 9234, negotiate and confirm the operator-intended relationship on each eBGP session using the BGP Role Capability and the Only-to-Customer attribute, providing in-band detection and prevention of route leaks [14]. When coupled with ROV enabled the RPKI architecture [25] [26], Roles help ensure that only legitimately authorized prefixes are propagated according to policy, reducing false positives in origin filtering and reinforcing secure import/export policies.

ASPA-based AS_PATH verification is more accurate and effective when aligned with confirmed BGP Roles, enabling granular route-leak mitigation and complementing both ROV and BGPsec's end-to-end path integrity guarantees.

Routing security incidents have increased in both frequency and severity over the years. The Internet community has developed various techniques to detect, mitigate, and prevent these incidents. Among these techniques, RPKI and ROV have gained the broadest adoption and have proven effective in safeguarding against route hijacking.

Nevertheless, persistent vulnerabilities necessitate new solutions. BGPsec and ASPA offer complementary validation and protection against various types of incidents. Consequently, network operators will likely need to master a combination of these techniques to significantly bolster their operational security.

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